SSD_Power Adjustment Reference

1. Turn off the analog circuit

Turn off the following options in the kernel config (Note: mark * to turn it off. If you need to click the screen to have UI display, DISP_LPLL/UPLL_0/UPLL_1 cannot be turned off; if you need to click on the mipi screen, MIPI_DPHY_TX cannot be turned off.)

```
[*]   Do not support debugfs for clock platform
[*]   Skip squashfs bad block
[*]   support deferred initcalls
[ ]     defer slab_sysfs_init
[ ]     defer param_sysfs_init
[ ]     defer perf_event_sysfs_init
[ ]     defer irq_sysfs_init, init_clocksource_sysfs,clockevents_init_sysfs
[ ]     defer of_core_init
[ ]   Do not create cryto test thread
[*]   SoC iNfinity2m (ARCH_MULTI_V7)
[ ]     iNfinity2m FPGA environment
[*]   Power down audio at boot
[*]   Power down EMAC at boot
[*]   Power down HDMI_ATOP at boot
[*]   Power down IDAC_ATOP at boot
[*]   Power down IDAC_LPLL at boot
[*]   Power down DISP_LPLL at boot
[*]   Power down MIPI_DPHY_TX_TOP at boot
[*]   Power down SATA_ATOP at boot
[*]   Power down UPLL_0 at boot
[*]   Power down UPLL_1 at boot
[*]   Power down USB20_P1 at boot
[*]   Power down USB20_P2 at boot
[*]   Power down USB20_P3 at boot
[ ]   Record timestamp in sram
```

2. Turn off the corresponding digital circuit

- The function modules corresponding to Audio & HDMI & IDAC(VGA) & DISP are configured in the SDK, and the corresponding SDK modules can be turned off:

  project\release\customer_tailor\nvr_i2m_display_glibc_tailor.mk

  - audio

    ```
    interface_ai:=disable
    interface_ao:=disable
    mhal_aio:=disable
    ```

  - hdmi

    ```
    interface_hdmi:=disable
    mhal_hdmitx:=disable
    ```

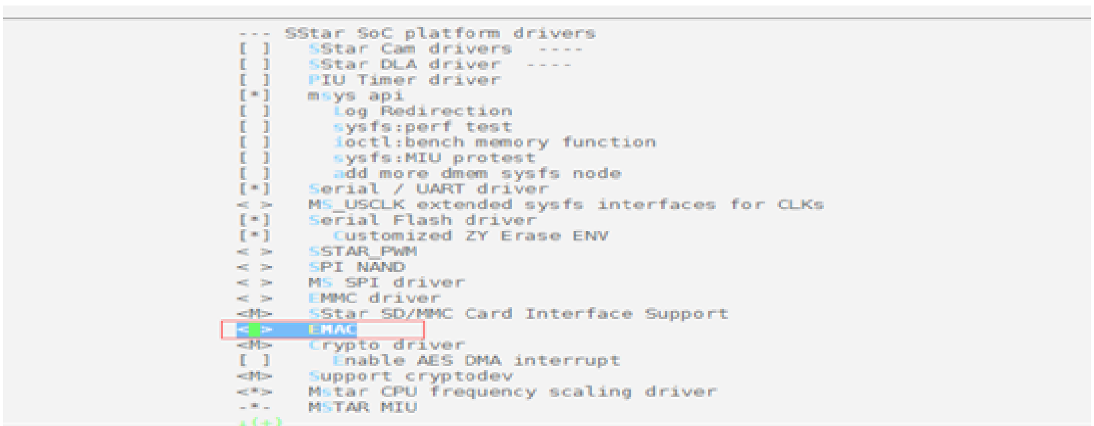
  - disp

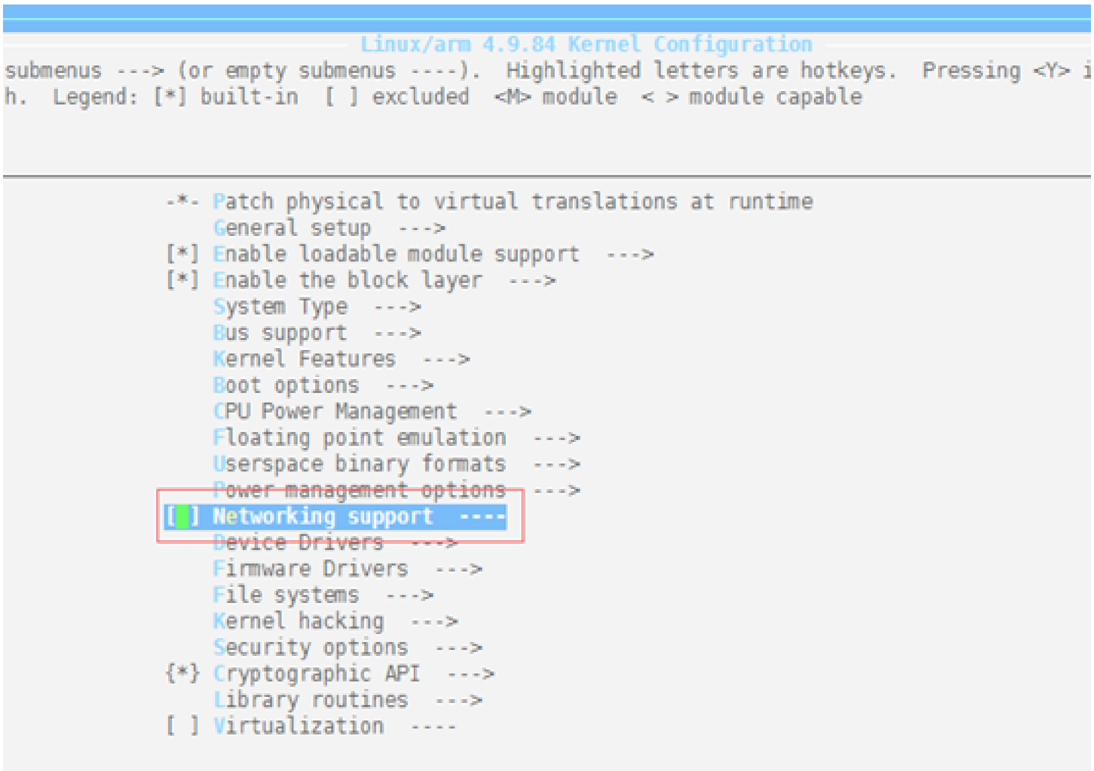
    ```
    interface_disp:=disable
    mhal_disp:=disable
    ```

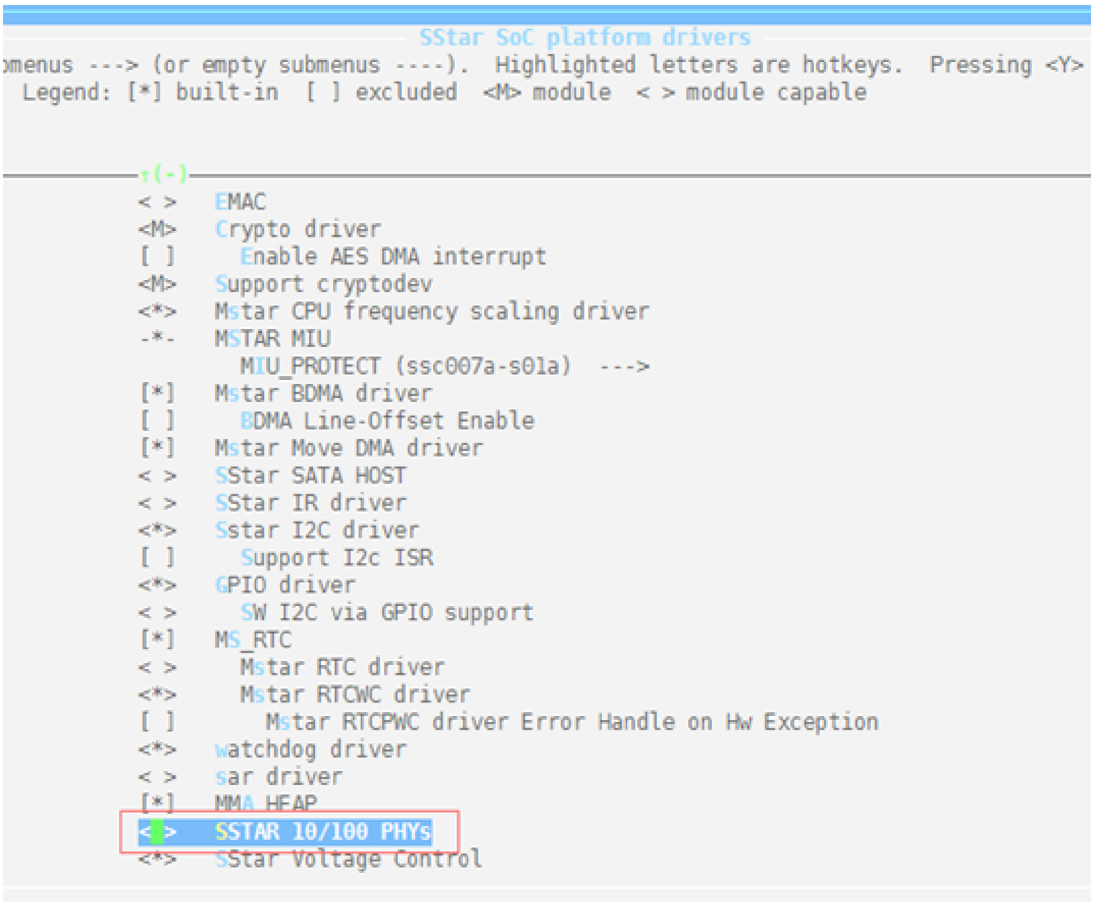
  - mipi tx

    ```
    interface_panel:=disable
    mhal_panel:=disable
    ```

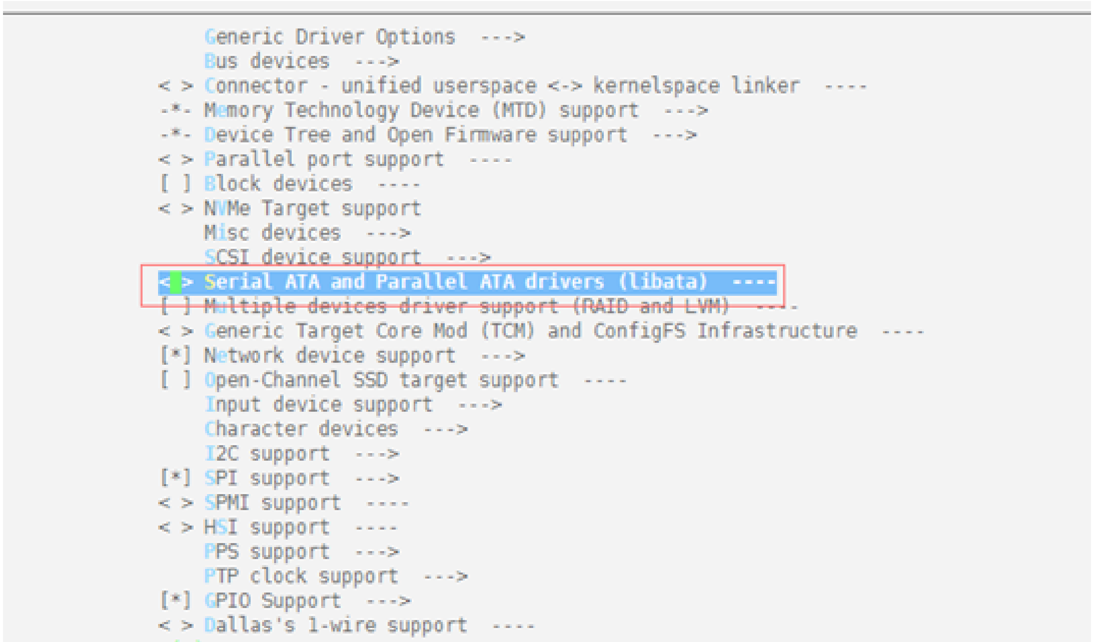
- EMAC网络开关的config在kernel的config里面：



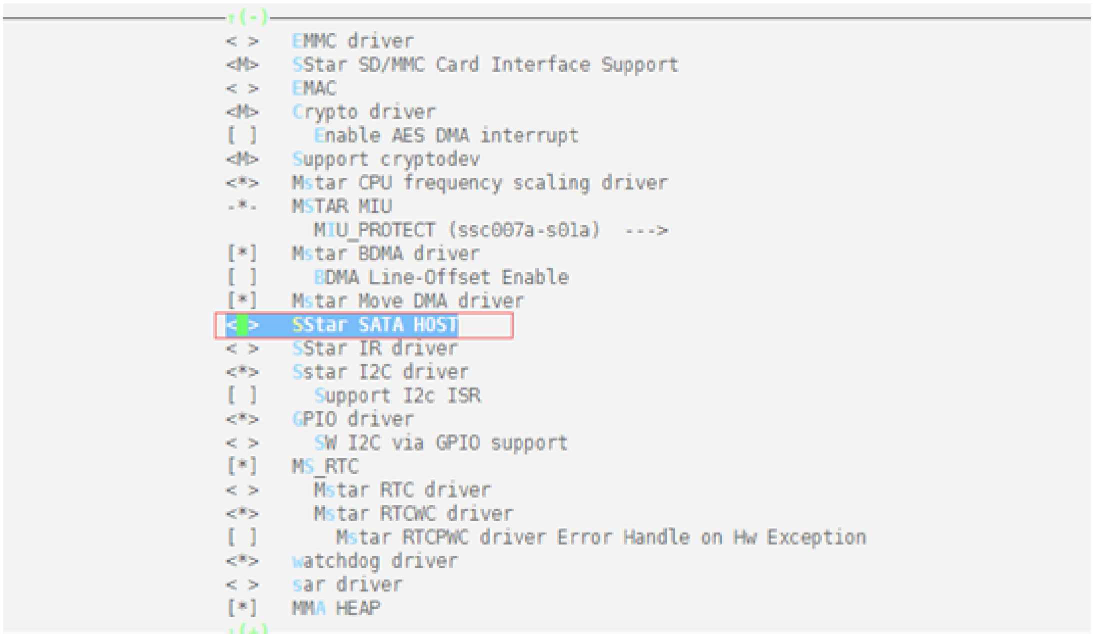




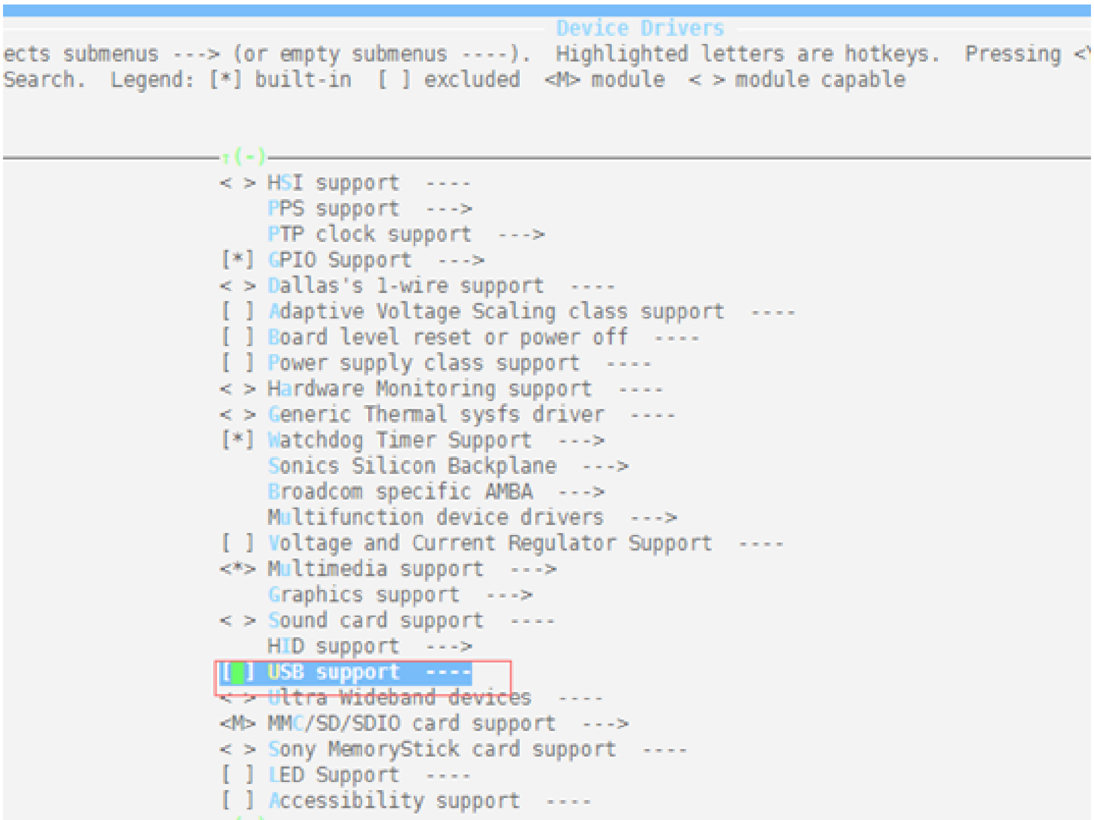
- Sata开关的config在kernel的config里面：



- SS_SATA_HOST:



- Usb开关的config在kernel如下config



3. CPU单核运行400MHz

- CPU 以400MHz运行

  kernel\arch\arm\boot\dts\infinity2m.dtsi

```
    cpus {
        #address-cells = <1>;
        #size-cells = <0>;

        cpu@0 {
            device_type = "cpu";
            compatible = "arm,cortex-a7";
            clock-frequency = <1000000000>;
            clocks = <&CLK_cpupll_clk>;
            reg = <0x0>;
            operating-points = <
                /* kHz     uV */
                /*
                1200000  1000000
                1100000  1000000
                1000000  900000
                */
                /*
                800000   900000
                600000   900000
                */
                400000   900000

            >;
        };
```

- CPU单核运行

  在bootargs最后加上nosmp即可：

  ```
  bootargs=console=ttyS0,115200 root=/dev/mtdblock4
  rootfstype=squashfs ro init=/linuxrc LX_MEM=0x3f00000

  mma_heap=mma_heap_name0,miu=0,sz=0x1000000
  mma_memblock_remove=1 highres=off

  mmap_reserved=fb,miu=0,sz=0x300000,max_start_off=0x3300000,max_en
   nosmp
  ```

公版的测试数据如下：

测试软件版本：TAKOYAKI_DLS00V009。

测试环境：Takoyaki NOR EVB板接TTL panel 12V供电。

		默认配置(网络&usb打开)	关闭网络	关闭网络 & usb	关闭网络 usb& sata
3.3V	IO	75.8	73	56.6	56.6
1.8V	AVDDIO_DRAM	32.1	32	32.02	32.02
	AVDDIO_DATA	49.5	50	49.4	49.5
0.9V	DVDD_DDR	7.54	7.5	7.52	7.52
	COREPOWER	127	124	120	120
	Total (unit: mA)	292	286	265.54	265.64

Check CPU temperature:

```
cat /sys/devices/virtual/mstar/msys/TEMP_R
```

Check CPU frequency:

```
cat sys/devices/system/cpu/cpu0/cpufreq/cpuinfo_cur_freq
```